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“Generalized Lymphadenopathy”

Differential diagnoses and principles of management

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Learning outcomes



At the end of this lecture, you will be able to (hopefully):

- List the differential diagnosis of enlarged lymph nodes
- Classify of major subtypes of lymphoma
- Understand common clinical presentations of lymphomas
- Understand the principles of investigations in lymphoma
- Understand the treatment principles of lymphomas
- List the major modalities of treatment of lymphomas



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Case scenarios

Case 1



- M/30 with good past health
- Complained of fever and rash for 3 days after hiking
- No recent drug / OTC use.
- P/E: generalized enlargement of LN (1-2 cm); liver edge and spleen tip palpable. Tonsils normal
- CBC showed WBC $7.78 \times 10^9 / L$ (N: 3.89-9.93); Hb 12.8g/dL (N: 13.3-17.1); Platelet count $115 \times 10^9 / L$ (N:167-396)
- LDH 350 U/L

Differential diagnoses of generalized lymphadenopathy



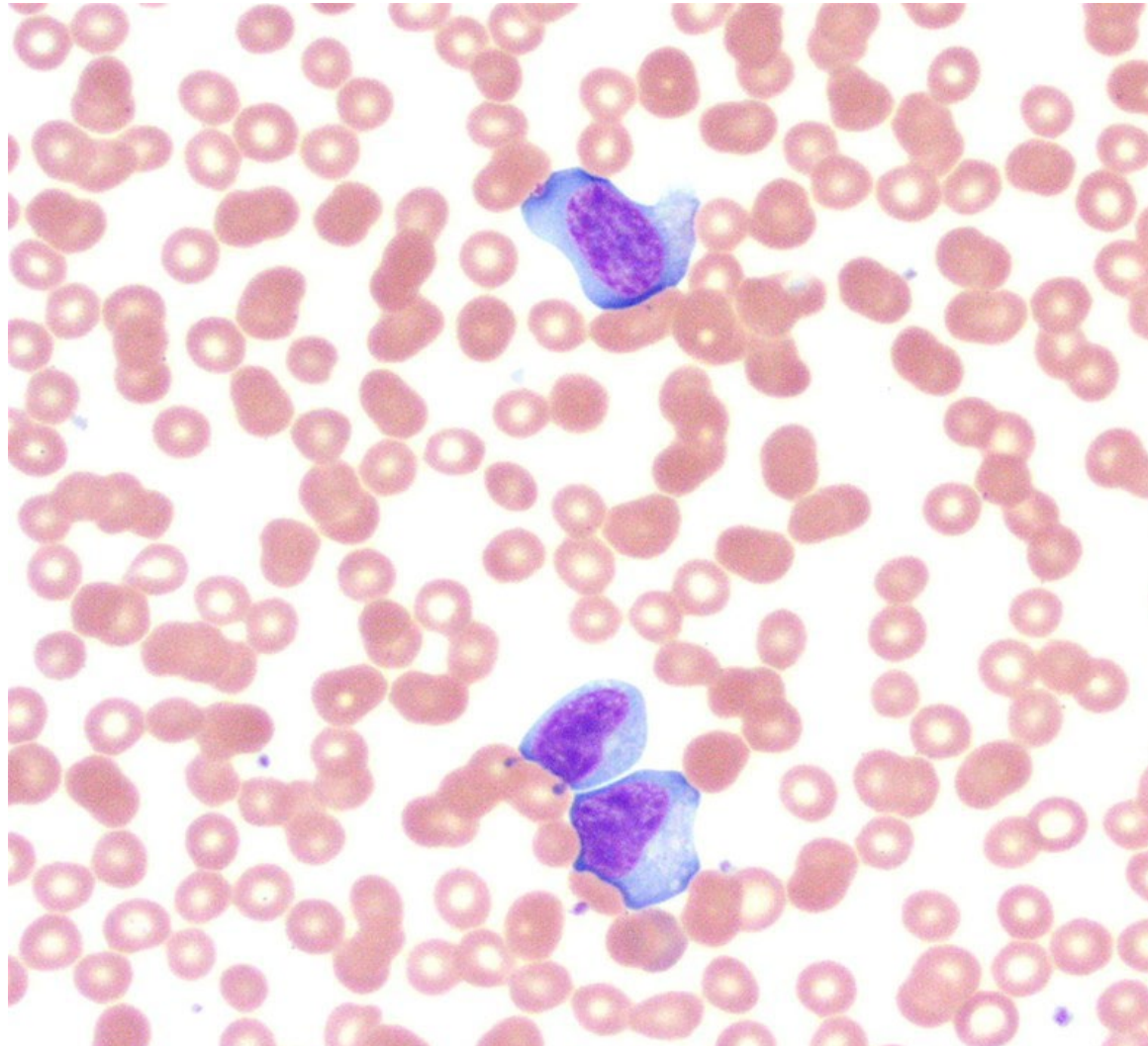
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- Neoplastic – leukaemias, lymphomas/LPD
- Infective (with predilection to involvement of RES) – Viral (EBV, CMV), bacterial, mycobacterial, parasitic, dimorphic fungi
- Autoimmune
- Drugs

Case 1 – peripheral blood film



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Differential diagnoses of atypical lymphocytosis / infectious mononucleosis-like syndrome



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- Viral infections (EBV, CMV, HIV, acute viral hepatitis)
- Bacterial (atypical organisms e.g. *Mycoplasma pneumoniae*, *Legionella pneumophila*, *Salmonella* sp., *Rickettsia* sp.)
- Mycobacterial (disseminated TB)
- Dimorphic fungi (*Talaromyces marneffe*)
- Parasitic (*Toxoplasma gondii*)
- Drug reactions / hypersensitivity

N.B. acute lymphoblastic leukaemia may mimic this (especially if the blood film is not carefully assessed)

Case 1



- Atypical lymphocytes on PB
- CXR clear
- Blood culture – no growth
- Sputum for C/ST, AFB smear negative
- Mycoplasma serology – non-reactive
- CMV IgG+ / IgM-
- CMVpp65 negative
- Monospot test and EBV serology negative
- Widal's test negative
- HIV 1/2 Ab negative
- Toxoplasma serology non-reactive
- Urine for Legionella Ag negative

Case 1



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- *Orientia tsutsugamushi* antibody titre: 256 on admission and 1024 at week 2
- Dx = scrub typhus
- (an eschar was actually present on his left inner thigh)

Case 2

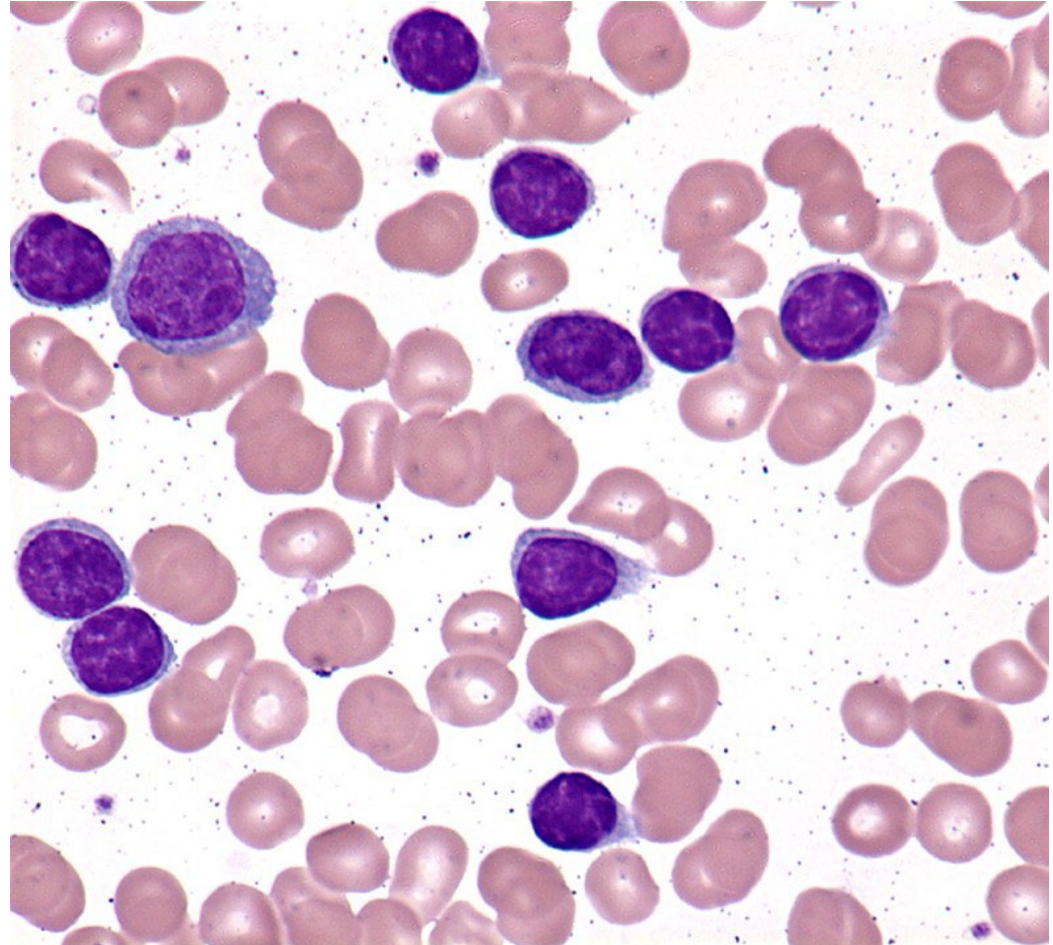
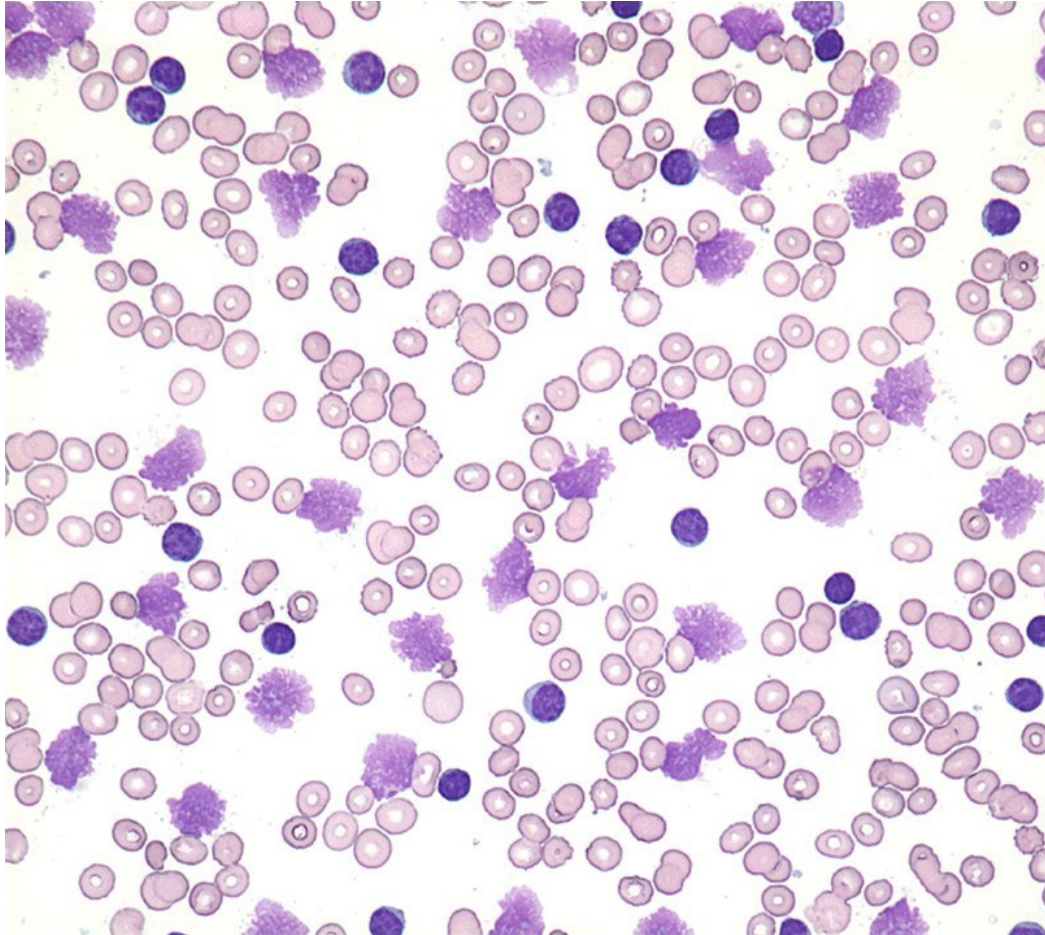


- M/62
- Complained of easy bruising for 2 week
- P/E: generalized enlargement of LN (2-3cm), palpable hepatosplenomegaly.
- CBC showed WBC $9.5 \times 10^9 / L$ (N: 3.89-9.93); Hb 12.2g/dL (N: 13.3-17.1); Platelet count $3 \times 10^9 / L$ (N:167-396)
- LDH 550 U/L

Case 2 – peripheral blood film



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Case 2



- Numerous “smear” or “smudge” cells
- Abnormal small-to-medium sized lymphoid cells with high N/C ratio, round nuclei, condensed chromatin, no nucleoli and smooth cytoplasmic border
- Little polychromasia and no spherocytes. No “rouleaux” formation
- Genuine thrombocytopenia with no clumps. Occasional giant platelets

Case 2



- Multiparametric flow cytometry – abnormal lymphoid cells light chain-restricted expressing CD19, CD5, CD23, CD20 (dim), surface Ig (dim), CD79b (dim). T-cell and NK cell markers negative
- BMA + trephine – heavy infiltration by abnormal lymphoid cells with the identical immunophenotype. Megakaryocytic hyperplasia.
- CG – normal karyotype
- FISH – del(17p) consistent with TP53 loss
- Platelet serology: Ab vs Gplb/IX and GpIib/IIIa +
- PET-CT – generalized LN and hepatosplenomegaly of “low grade” SUVmax uptake

Case 2



- CLL - Binet stage C
- Adverse risk due to *TP53* loss
- Started ibrutinib (BTK inhibitor) with initial response but progressed after 6 months
- Did not response – given salvage treatment with obinutuzumab and venetoclax (bcl-2 inhibitor) and achieved CR

Case 3

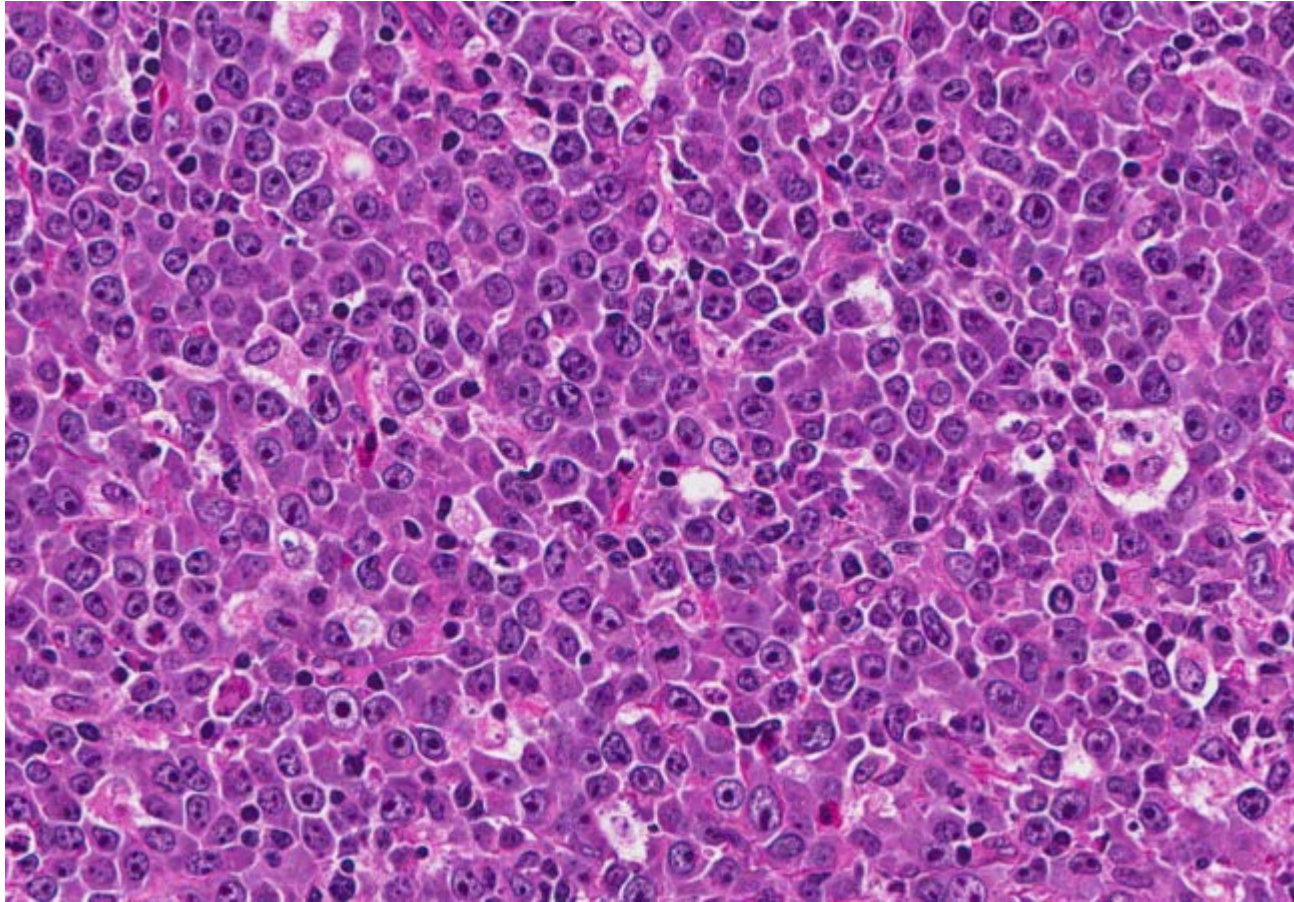


- A 63 year-old man presented with weight loss and rapidly enlarging bilateral cervical lymph nodes for 1 month
- He had history of hypertension since the age of 50 that was well-controlled with one anti-hypertensive
- Physical examination showed bilateral multiple firm and non-tender cervical lymph nodes of 3-5cm in size.
- CBC, LRFT and serum electrolytes were normal
- LDH and urate levels were grossly elevated

Diagnostic investigation - histology



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1. Effaced nodal architecture
2. Sheets of large abnormal lymphoid cells
3. Immunohistochemistry showed CD19+, CD20+, CD79a+, PAX5+. T-cell and NK cell markers were negative

Consistent with **DIFFUSE LARGE B-CELL LYMPHOMA**

Case 3 cont'd



- PET-CT showed hypermetabolic LN in bilateral cervical, thoracic and intraabdominal regions
- BM biopsy – Not involved by lymphoma
- Ann-Arbor Stage IIIB disease
- ECG and Echocardiogram showed normal findings
- CXR was clear
- HBsAg positive, anti-HCV negative, anti-HIV 1/2 ab negative
- HBV DNA undetectable
- G6PD normal

Case 3 cont'd

- Patient started on entecavir prophylaxis
- DLBCL was treated with R-CHOP
- R=Rituximab; C=cyclophosphamide;
H=doxorubicin (hydroxyl-daunorubicin);
O=vincristine (Oncovin); P=prednisolone

What are the mechanisms of action and side-effects of each of the drugs in the regimen R-CHOP?

Case 3 cont'd



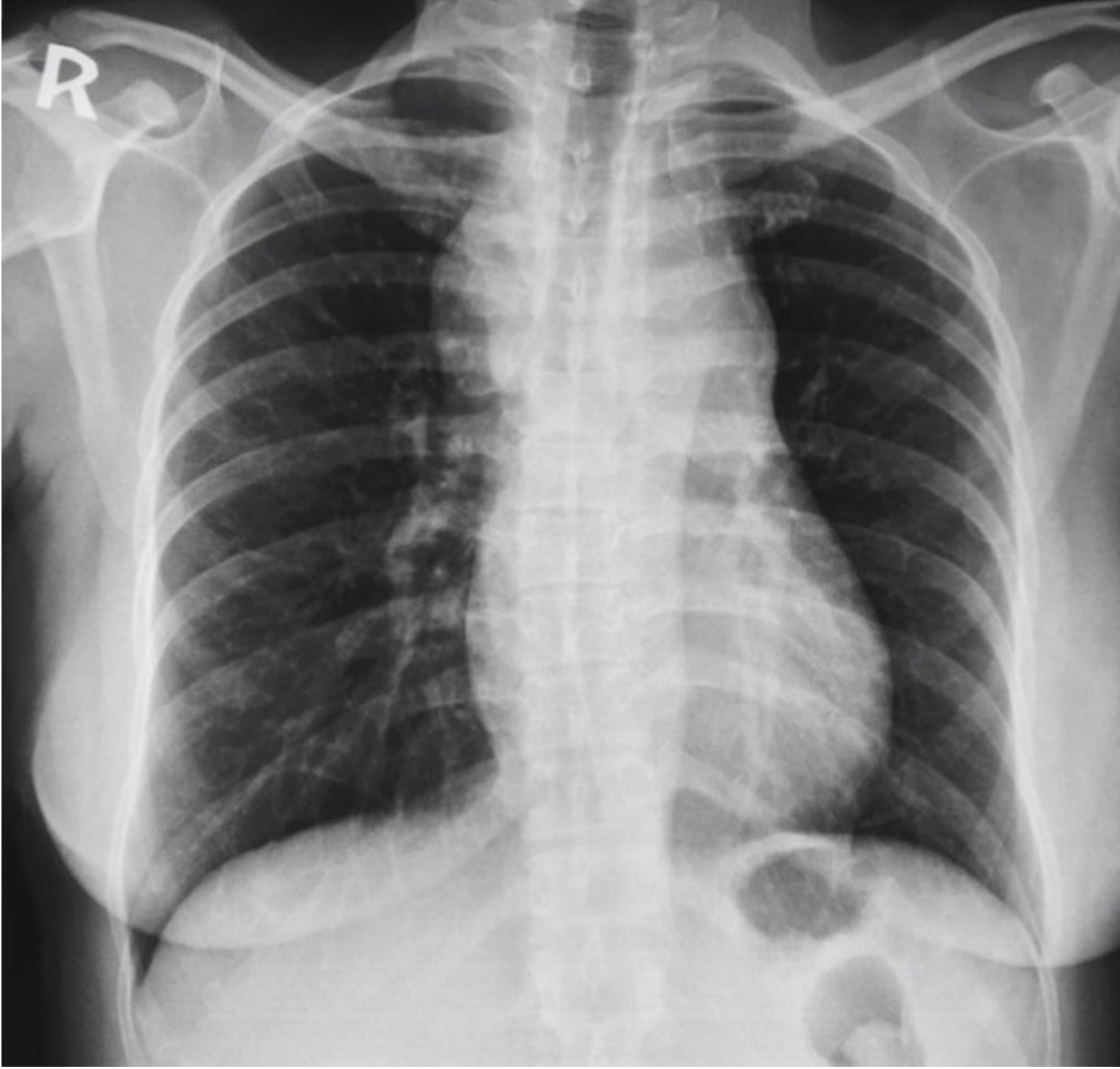
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- Patient achieved complete response (CR) after 3 cycles of R-CHOP
- Completed 6 cycles of R-CHOP and remained in CR

Case 4



- A 35 years-old woman presented with dry cough for 2 months associated with fever and night sweats. Recent loss of appetite and weight loss of 10kg over 2 months
- She was a non-smoker with good past health
- Physical examination showed a 3-cm palpable left supraclavicular fossa lymph node that was rubbery in consistency.
- The liver and spleen were not palpable.
- CVS and Resp exam were unremarkable
- No upper limb or facial swelling



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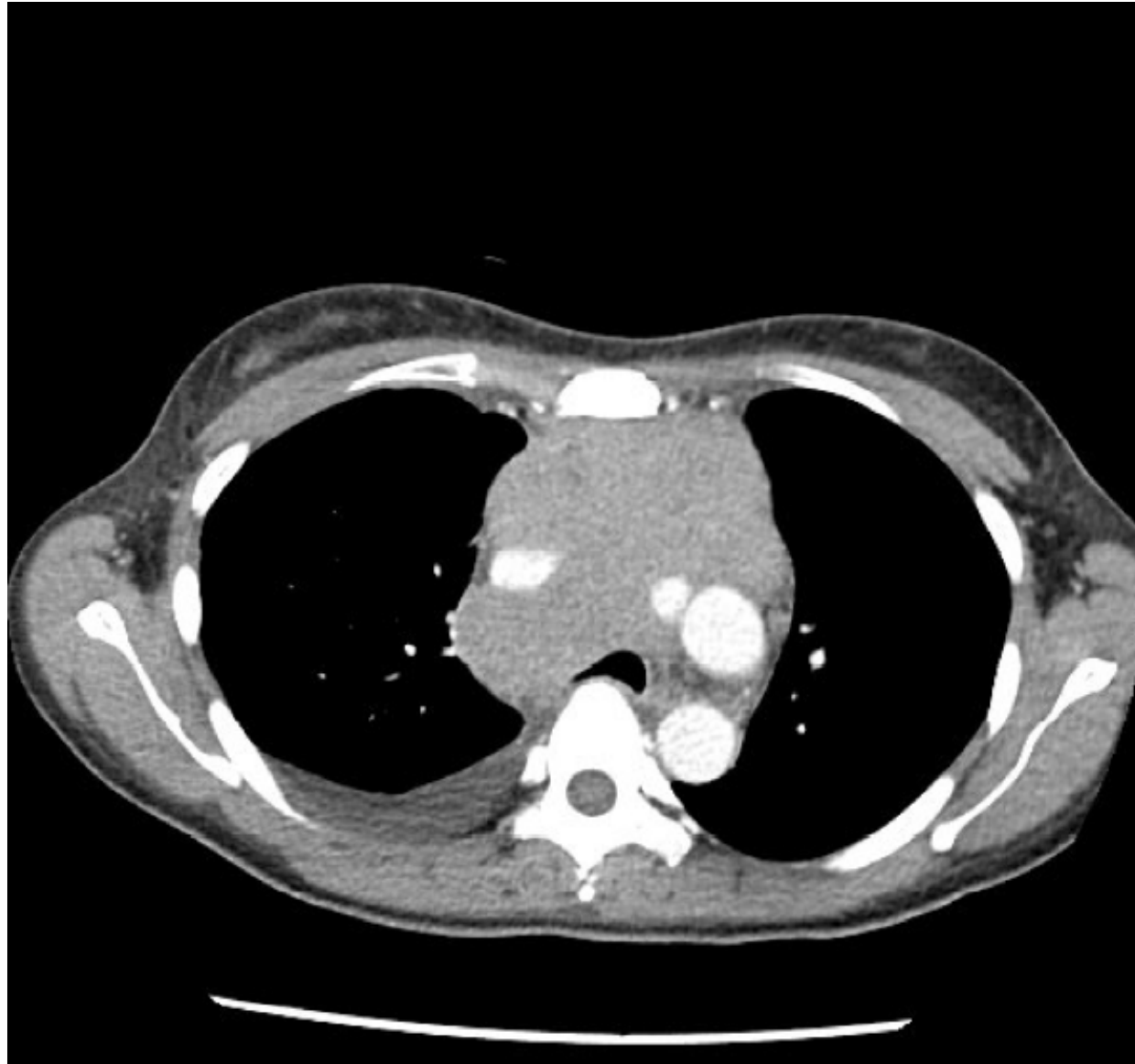
Mediastinal mass Differential diagnoses:

- 1. LN – lymphoma / metastatic LN**
- 2. Thymoma**
- 3. Germ cell tumour**
- 4. Others – retrosternal goitre, dilated aortic arch, neurofibroma**

Case 4 – CT thorax



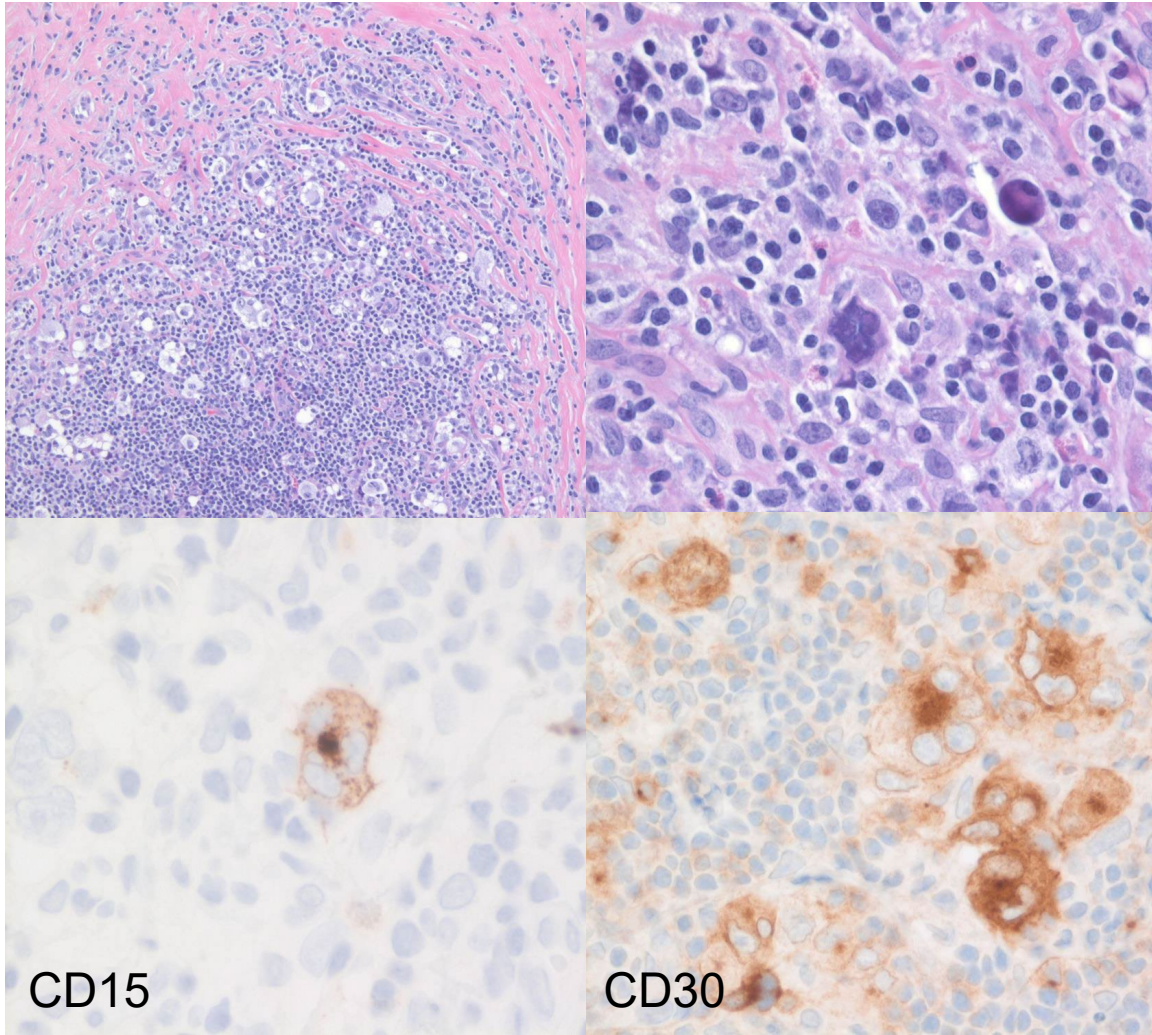
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Diagnostic investigation - histology



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Excisional biopsy of L SCF LN:

1. Distorted nodal architecture with thick fibrotic bands
2. Typical multinucleated Reed-Sternberg cells were present
3. Prominent infiltration by inflammatory cells (reactive lymphocytes, eosinophils and plasma cells)
4. CD20 negative, CD15+, CD30+, T- and NK-cell markers negative

Case 4 cont'd



- Dx = Nodular Sclerosis Classical Hodgkin Lymphoma
- Ann Arbor Stage IIB
- CBC showed eosinophilia
- LRFT was normal, serum electrolytes were normal
- LDH and urate levels were elevated
- Baseline ECG, echocardiogram, lung function studies were normal
- HBsAg negative, anti-HBs positive, anti-HBc negative, anti-HCV ab negative, anti-HIV 1/2 ab negative
- Patient underwent oocyte preservation before starting chemotherapy with Adriamycin, Bleomycin, Vinblastine and Dacarbazine (ABVD)

Case 4 cont'd

- Cycle 1 ABVD started with adequate hydration and febuxostat (xanthine oxidase inhibitor)
- CBC and LRFT monitored regularly in between cycles
- No cardiac or resp symptoms during treatment
- Interim assessment PET-CT after 2 cycles of ABVD showed complete response (CR)
- Complete 4 cycles of ABVD and end-of-treatment PET-CT showed continual remission

What are the mechanisms of action and side-effects of each of the drugs in the regimen ABVD?

Case 4 cont'd



- Unfortunately, the patient had relapsed disease 8 months after remission with nodal involvement at bilateral cervical, mediastinal, and intraabdominal lymph nodes on PET-CT
- Re-biopsy of the cervical LN confirmed recurrent Classical Hodgkin Lymphoma, nodular sclerosis type.
- This patient was started on the immuno-conjugate Brentuximab vedotin (anti-CD30 monoclonal ab linked to MMAE)

Case 4 cont'd



- After 3 cycles of Brentuximab vedotin, the patient failed to achieve a response
- This patient was started on the check-point inhibitor pembrolizumab (anti-PD1) as salvage treatment

Case 4 cont'd



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- The patient achieved complete response after 4 cycles of pembrolizumab and treatment was continued



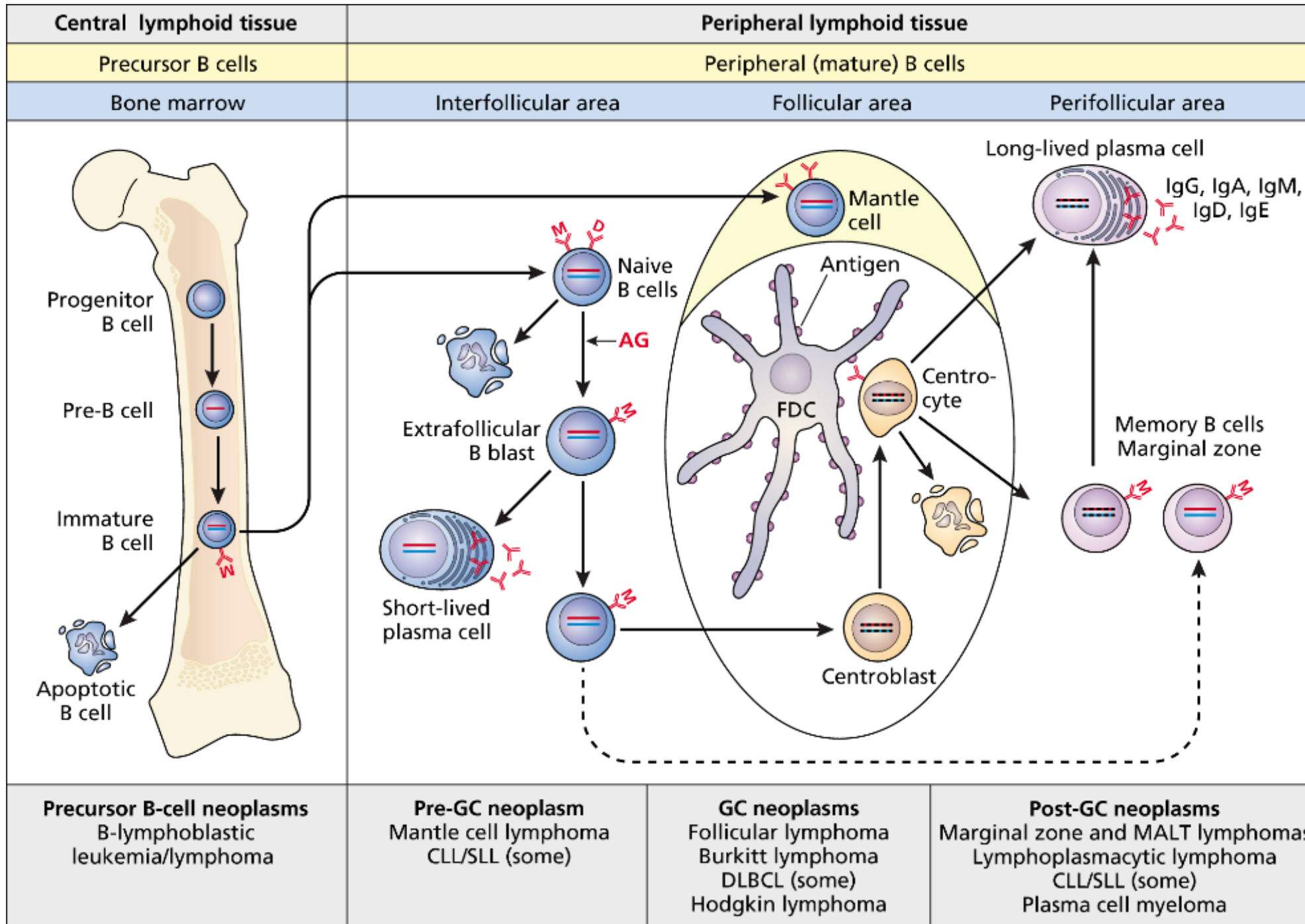
Principles of diagnosis, investigations and management of lymphomas

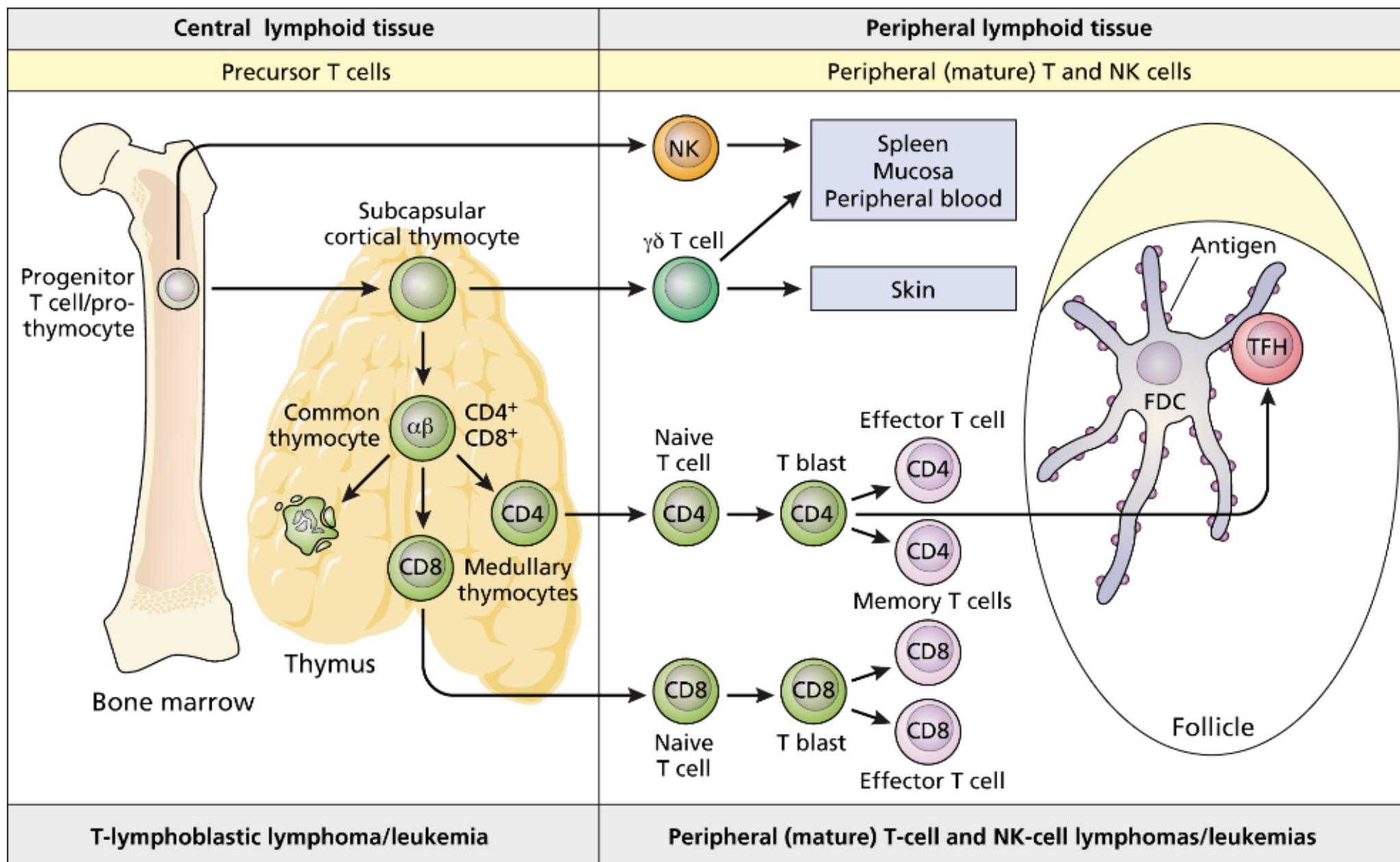
Introduction – a simplified definition



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- Lymphoma = neoplasm of the lymphoid system (lymphoid organs/ cells of B-, T- or NK- origin)
- In general, lymphoma usually refers to mature lymphoid neoplasm
- The accurate diagnosis of lymphoma requires integration of clinical, morphologic, immunophenotypic, and genetic features

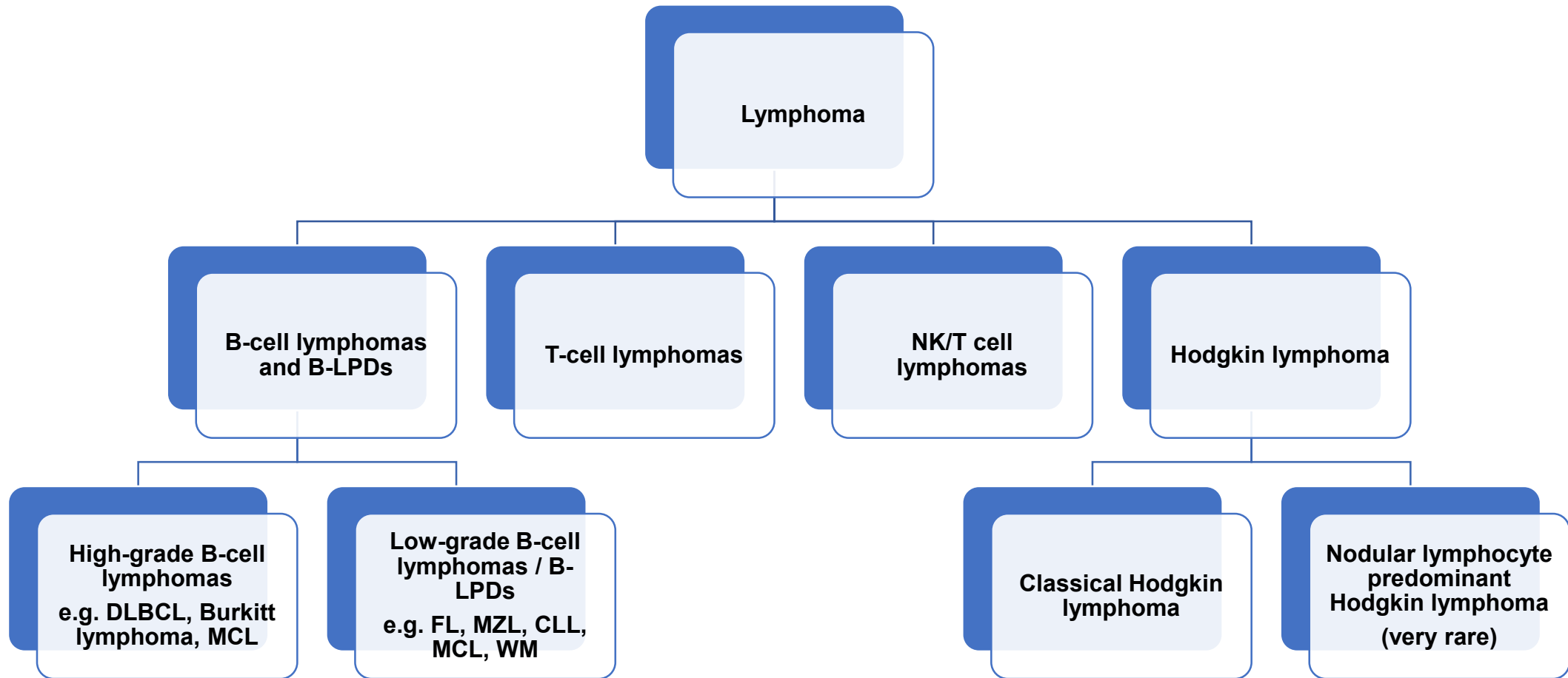




A practical classification of lymphoma



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Don't attempt to recite the WHO classification – it is beyond your scope !

Risk factors in the development of lymphomas



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Viral infections

- EBV, HTLV-1, HHV-8, HCV

Bacterial infection

- Helicobacter pylori
- Chlamydia psittaci

Impaired/altered immunity

- Congenital immunodeficiency
- Acquired immunodeficiency - HIV, organ or stem cell transplantation, aging, chronic immunosuppressive drugs
- Autoimmune - RA, SLE, Sjögren syndrome, celiac disease

Principles of diagnosis of lymphomas



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- Accurate history taking and physical examination
- Proper tissue diagnosis
- An adequate biopsy of the representative site is essential for histologic, immunohistochemical and genetic assessment
- Simple cytological assessment (e.g. fine-needle aspirations) are NOT acceptable

Investigations for lymphomas - aims



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1. Accurate diagnosis
2. Staging
3. Detect complications of lymphoma
4. Determine prognosis
5. Determine suitability or fitness for treatment
6. Detecting complications of treatment
7. Assessing response to treatment

Investigations in lymphomas



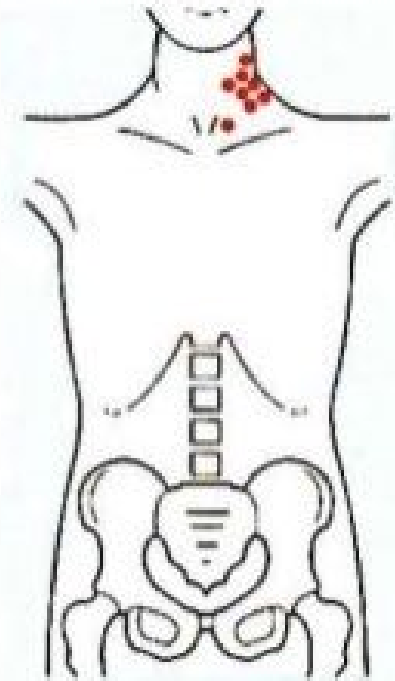
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- Complete blood count
- Liver and renal function tests
- Serum electrolytes (Na, K, Ca, PO₄)
- LDH and urate levels
- CXR (to look for disease-related complications, active or prior infections)
- ESR (prognostic marker for early stage cHL)
- SPE
- β 2-microglobulin (prognostic marker for follicular lymphoma)
- Bone marrow aspiration and trephine biopsy (for staging)
- Whole body 18-FDG positron emission tomography- computerized tomography (PET-CT) (for staging)

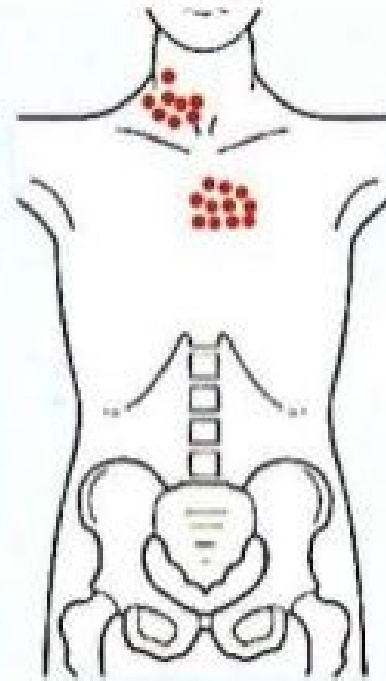
Staging of lymphomas – Ann Arbor Staging



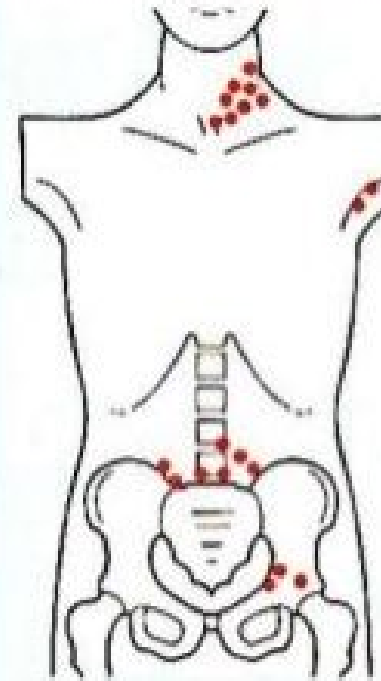
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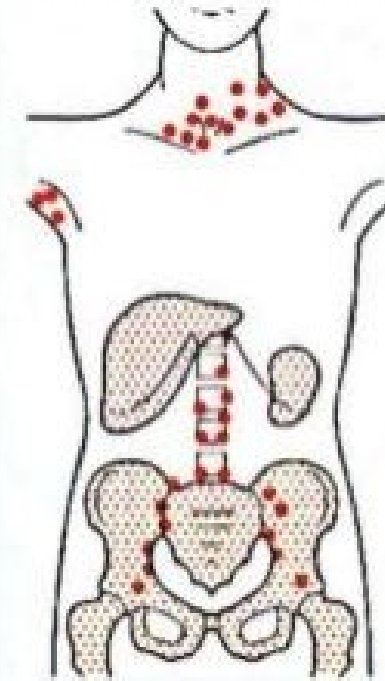
Stage I:
involvement of single lymph node region or single extralymphatic site (I_E)



Stage II:
involvement of two or more lymph node regions on same side of diaphragm; may include localized extralymphatic



Stage III:
involvement of lymph node regions on both sides of the diaphragm; may include spleen (III_S) or localized



Stage IV:
diffuse extralymphatic disease (e.g. in liver, bone marrow, lung, skin)

Investigations



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- ECG and transthoracic echocardiogram (risk of cardiotoxicity with anthracyclines)
- Lung function studies (risk of pulmonary fibrosis with bleomycin)
- Hepatitis B and C serology (risk of reactivation upon immunosuppression)
- HIV serology (Lymphomas may be a presenting feature of HIV AIDS)
- G6PD assay (risk of oxidative hemolysis with Co-trimoxazole)

Principles of management of lymphomas



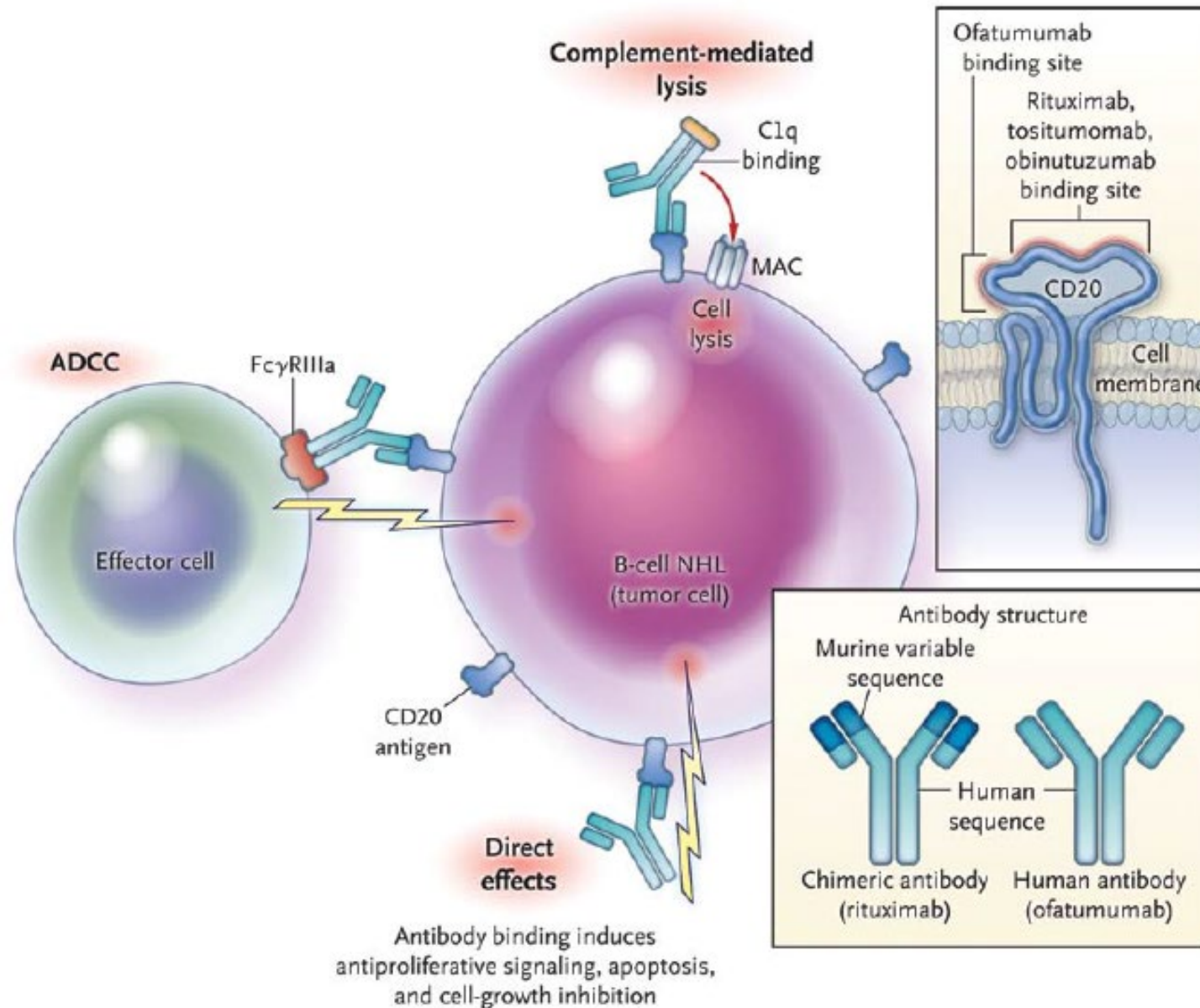
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- Multi-agent cytotoxic chemotherapy +/- monoclonal antibodies (e.g. anti-CD20 in CD20+ B-cell lymphomas)
- Targeted therapy (BTK or BCL-2 inhibitors) (e.g. in CLL)
- Multiple cycles of chemotherapy with interim and end-of-treatment assessment by imaging +/- biomarkers
- The use of immuno-conjugates, checkpoint inhibitors, bi-specific antibodies for certain subtypes of lymphoma
- Supportive care with the prevention of tumour lysis syndrome and management of disease/treatment complications
- HSCT (autologous HSCT, in general) is reserved for selected relapsed patients who are chemo-sensitive
- Chimeric antigen receptor T (CART) cell therapy for specific subtypes of B lymphomas (e.g. R/R DLBCL)

Monoclonal antibody against CD20



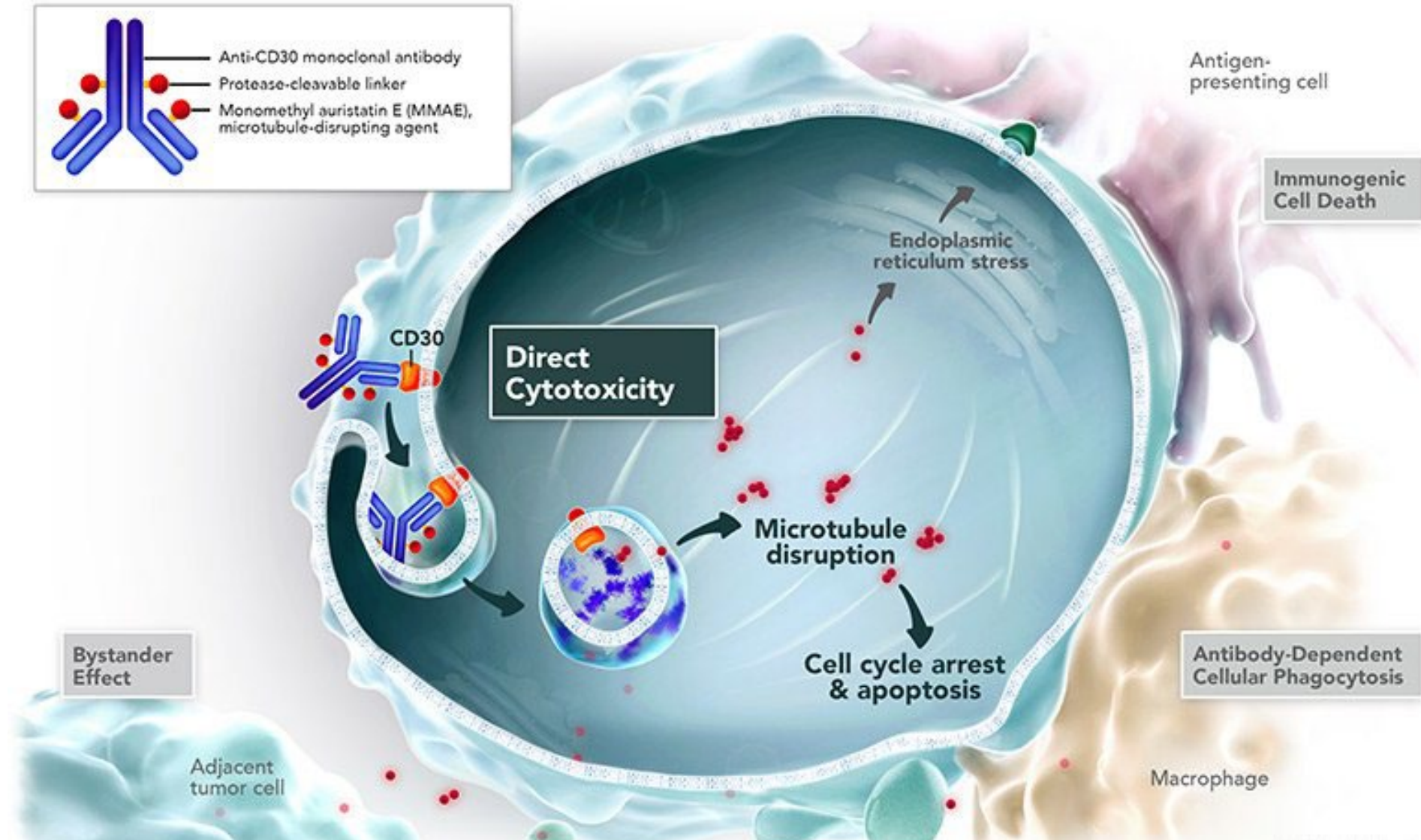
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Anti-CD30 immunoconjugate



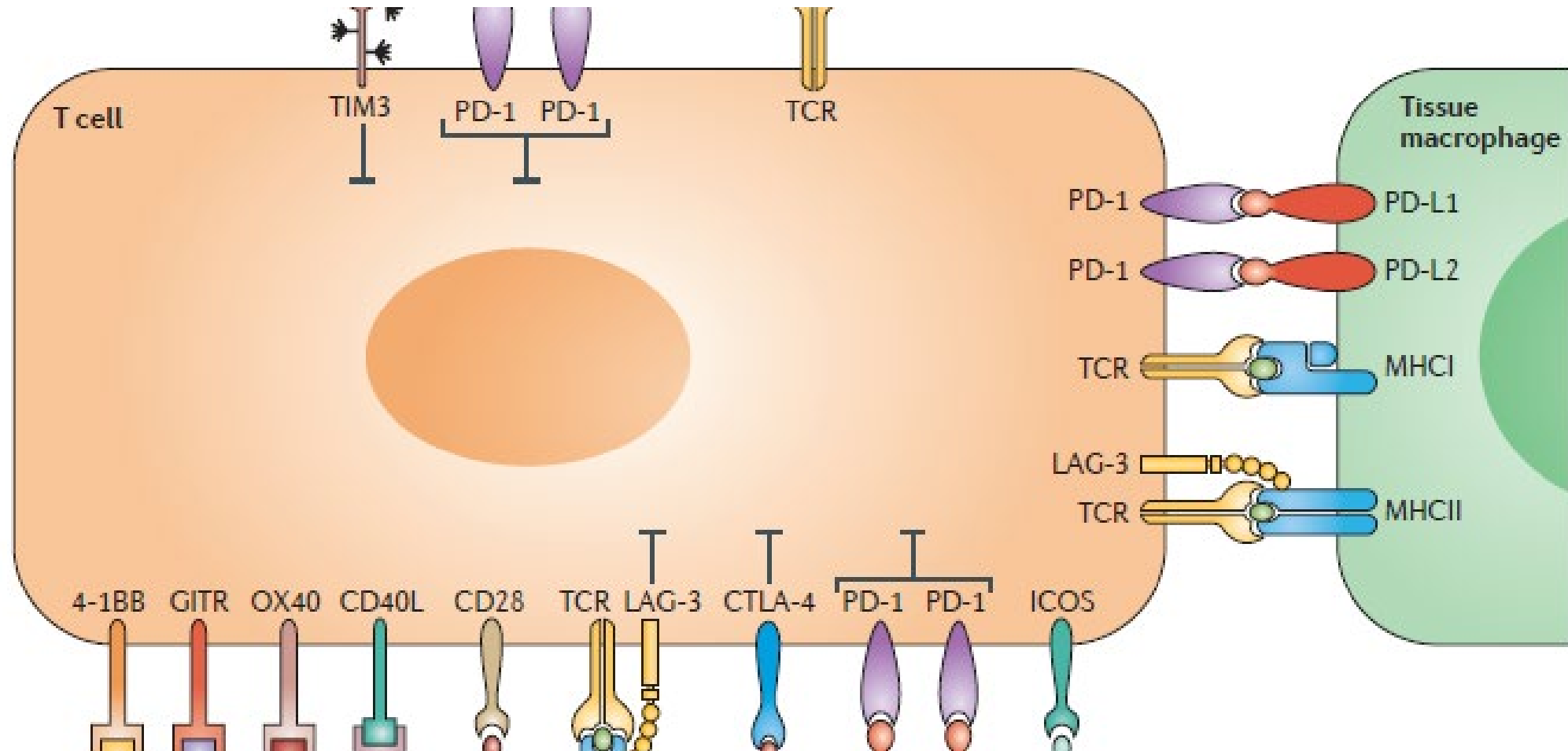
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Immune check-points and their inhibition



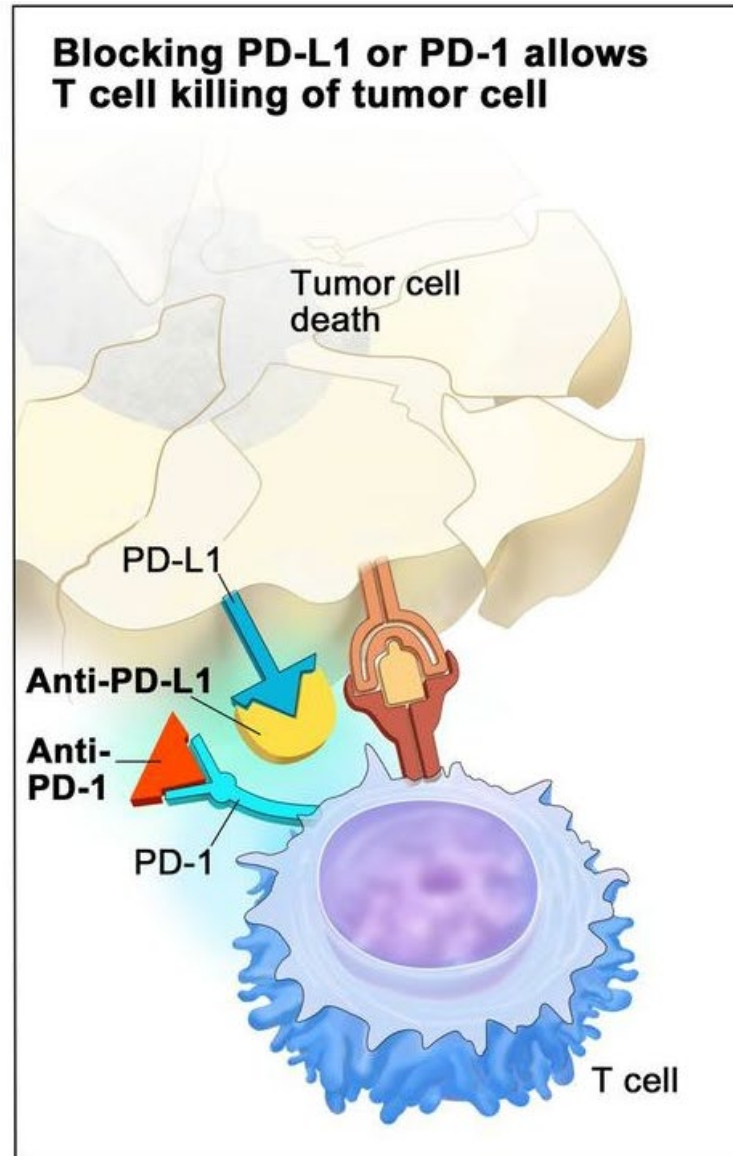
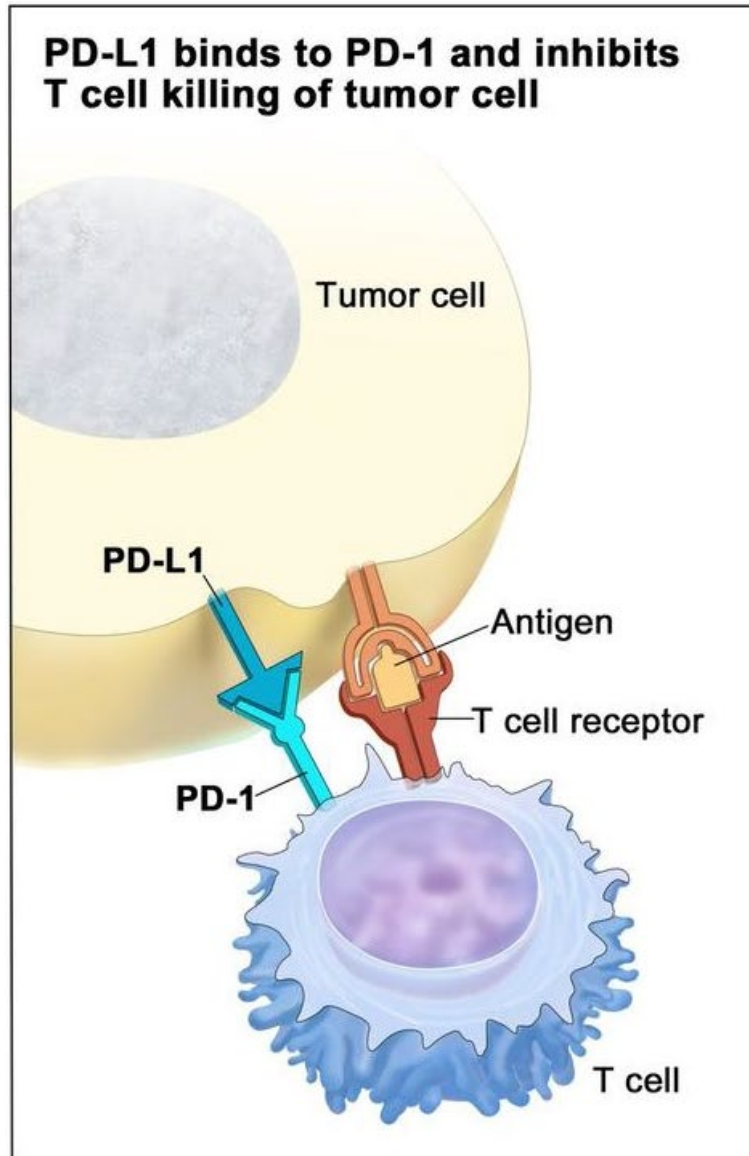
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PD-L1 and PD-1 inhibition



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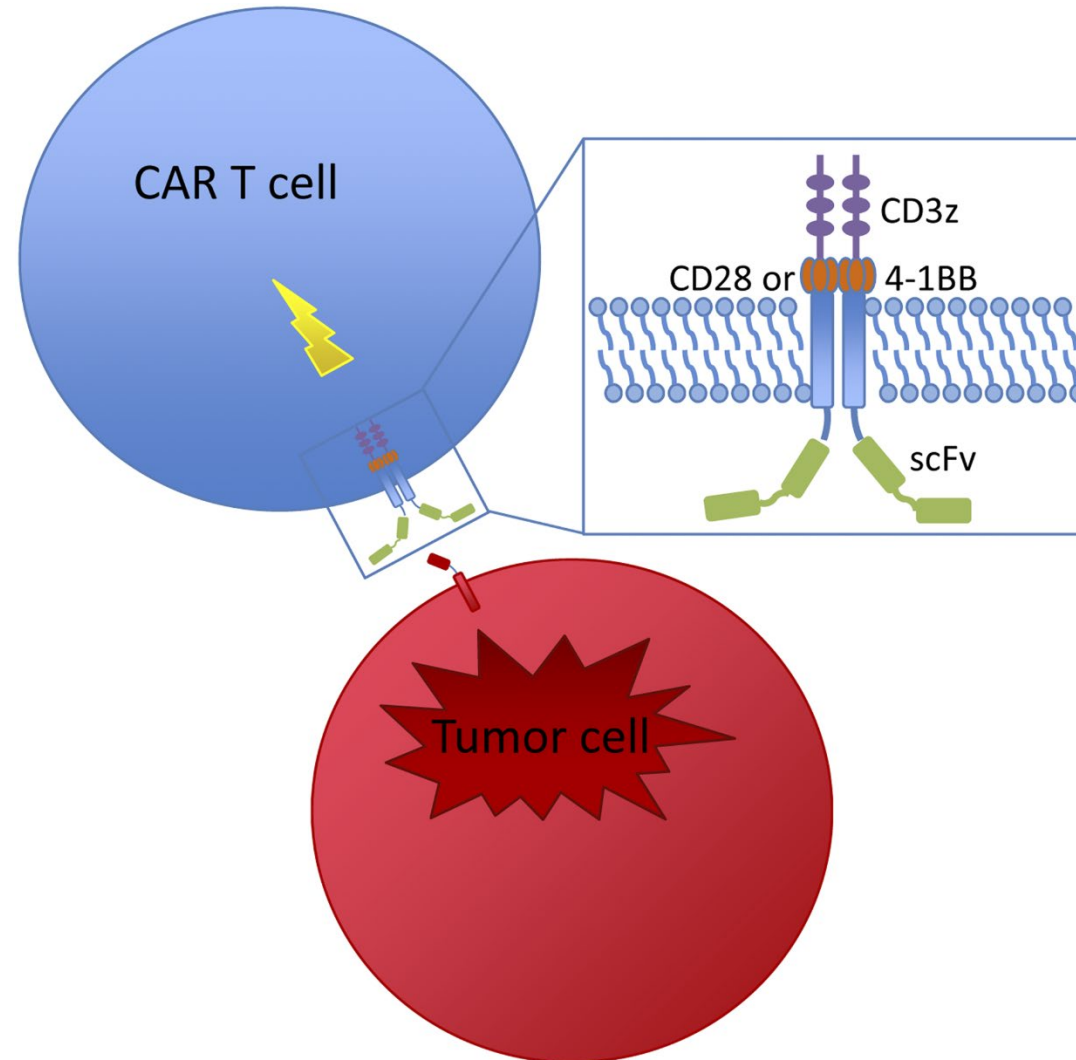
Chimeric antigen receptor (CAR) T-cell therapy



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- Adoptive cell therapy with T cells genetically engineered to express a CAR
- CAR is a recombinant receptor construct consisting of:
 1. extracellular single-chain variable fragment of an antibody recognizing a tumor-associated cell surface antigen;
 2. a spacer or hinge region, and;
 3. the TCR CD3 ζ chain

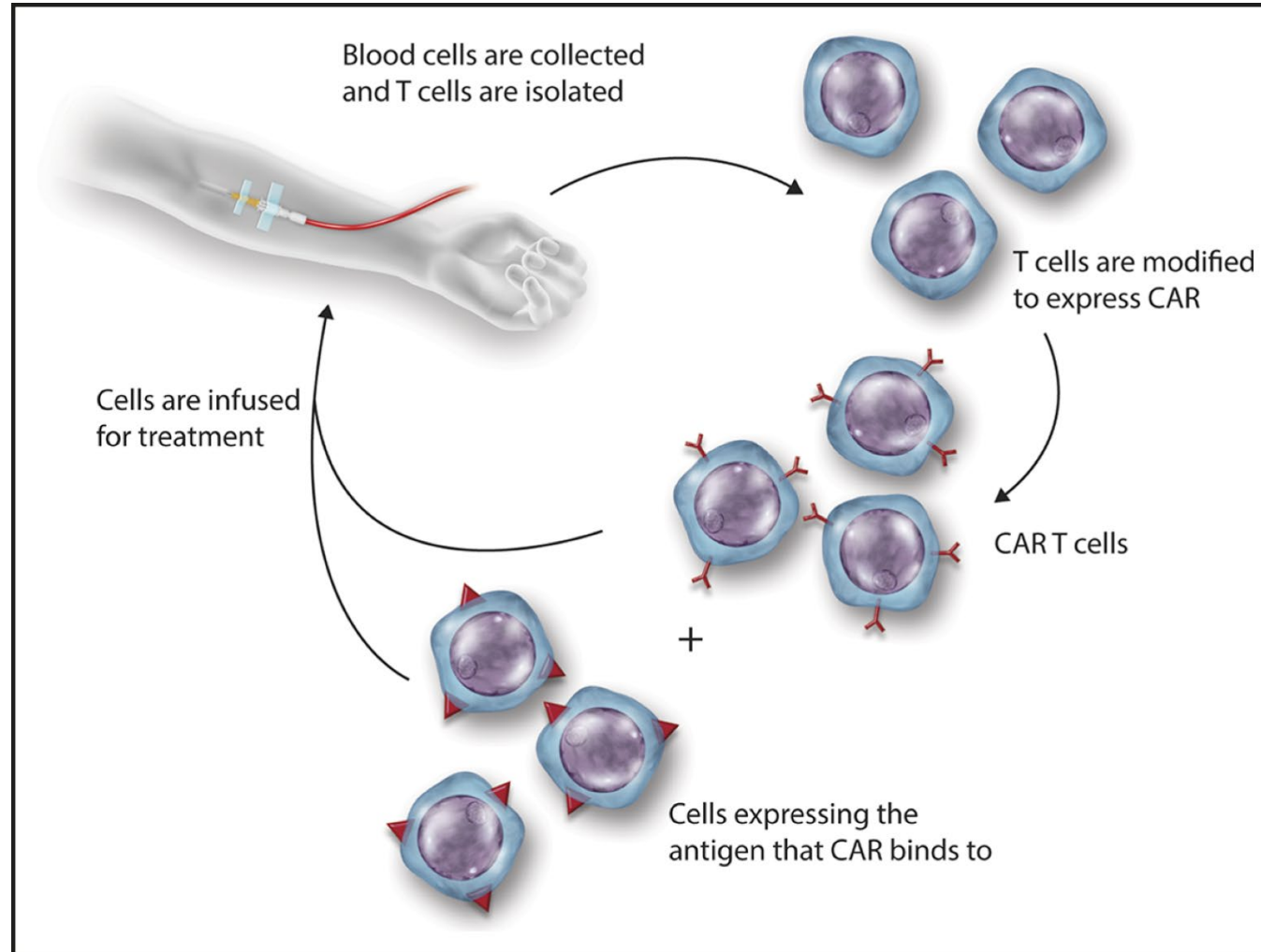
CAR-T cell therapy principles



CAR-T cell therapy principles



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Important subtypes of lymphomas and lymphoproliferative disorders

Hodgkin Lymphoma



- 10% of all lymphoma cases
- bimodal age distribution with one peak in the early 20s and the second in the mid-70s.
- Slight male predominance
- Usually presents with painless lymphadenopathy involving the neck and chest.
- Systemic symptoms of fevers, night sweats, and unexplained weight loss may occur in patients with advanced-stage disease
- majority of patients cured with combination chemotherapy with or without radiation

Hodgkin lymphoma - pathology



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- Monoclonal lymphoid neoplasm derived from B cells
- Two distinct entities, classical Hodgkin lymphoma (cHL) (95% of cases) and nodular lymphocyte-predominant Hodgkin lymphoma (NLPHL)
- Within classical cHL, there are four histologic subtypes: nodular sclerosis (NS), mixed cellularity (MC), lymphocyte rich (LR), and lymphocyte depleted (LD).
- Hallmark of cHL =Hodgkin Reed-Sternberg (HRS) cell
- NS cHL: 70% of cases; M/F equally affected; Typical age of presentation of NS cHL=15-35
- Mediastinal involvement, which may be bulky, is more common in NS cHL

Diffuse large B-cell lymphoma (DLBCL)



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- DLBCL is composed of large B cells with a diffuse growth pattern
- Subclassification – molecular (GCB / ABC); pathologic; defined disease entities (e.g. primary CNS DLBCL, primary mediastinal large B-cell lymphoma)
- 30% of all NHLs
- present with nodal (more common) or extranodal disease.
- BM involvement <10% of cases

DLBCL con't



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- In general, approximately two-thirds of patients are cured
- Prognostic factors – clinical (IPI), biologic (genetic subtype, MYC rearrangement)
- Backbone of treatment is R-CHOP
- Novel approaches for adverse biologic subtypes (MYC and BCL2 and/or BCL6)
- Auto-HSCT for chemosensitive relapse
- Allogeneic HSCT not routinely done
- CAR T-cell therapy for R/R CD19+ DLBCL

Burkitt lymphoma (BL)



- Most aggressive of all human malignancies
- Rapid doubling time, acute onset and progressive
- Deregulated high-level expression of cMYC arising from reciprocal translocation with immunoglobulin-heavy t(8;14) or variant light-chain gene loci t(2;8) or t(8;22)
- Frequently present with a bulky abdominal mass, B-symptoms and extranodal disease, including BM involvement (up to 70%).
- Leptomeningeal involvement, may be present at diagnosis in up to 30% of patients

Burkitt lymphoma



- Urgent treatment required
- Prevention of tumour lysis syndrome
- Rituximab + multi-agent chemotherapy
- Good prognosis with cure in $> 80\%$

Follicular lymphoma



- Prototypical and most common indolent lymphoma
- Limited-stage (Ann Arbor I or II) FL is relatively uncommon
- Most patients diagnosed at advanced stage
- Prognosis favourable with anti-CD20 + purine analogue/alkylating agent based therapy

Marginal zone B-cell lymphoma



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- 3 subtypes: extranodal MZL of MALT type, nodal MZL, and splenic MZL (SMZL)
- Associated with chronic antigenic stimulation by microbial pathogens or autoantigens: gastric MALT (*H pylori*), cutaneous MALT (*B burgdorferi* or *afzelii*), ocular adnexal MALT (*C psittaci*), nodal MZL (hepatitis C), SMZL (hepatitis C), pulmonary or parotid MALT (Sjögren syndrome), and thyroid MALT (Hashimoto thyroiditis).

Lymphoplasmacytic lymphoma and Waldenström macroglobulinaemia



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- LPL = Indolent neoplasm of small B lymphocytes, plasmacytoid lymphocytes, and plasma cells.
- Waldenström macroglobulinemia = presence of a monoclonal IgM gammopathy in the blood and clonal lymphoplasmacytic cells in BM
- IgM monoclonal protein in the blood: symptoms secondary to hyperviscosity, cryoglobulinemia, bleeding disorders, autoimmune hemolytic anemia)
- BM or tissue infiltration by the lymphoplasmacytic cells: anemia, hepatosplenomegaly, lymphadenopathy
- Autoimmune phenomena driven by the monoclonal protein: polyneuropathy

Mantle cell lymphoma



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- 6% of all NHLs
- Median age in the mid 60s
- striking male predominance
- strong tendency to present with advanced-stage disease.
- Extranodal involvement is common, including BM and PB,
- Peculiar tendency to invade the GI tract (“lymphomatous polyposis” of the large bowel)

Chronic lymphocytic leukaemia / small lymphocytic lymphoma



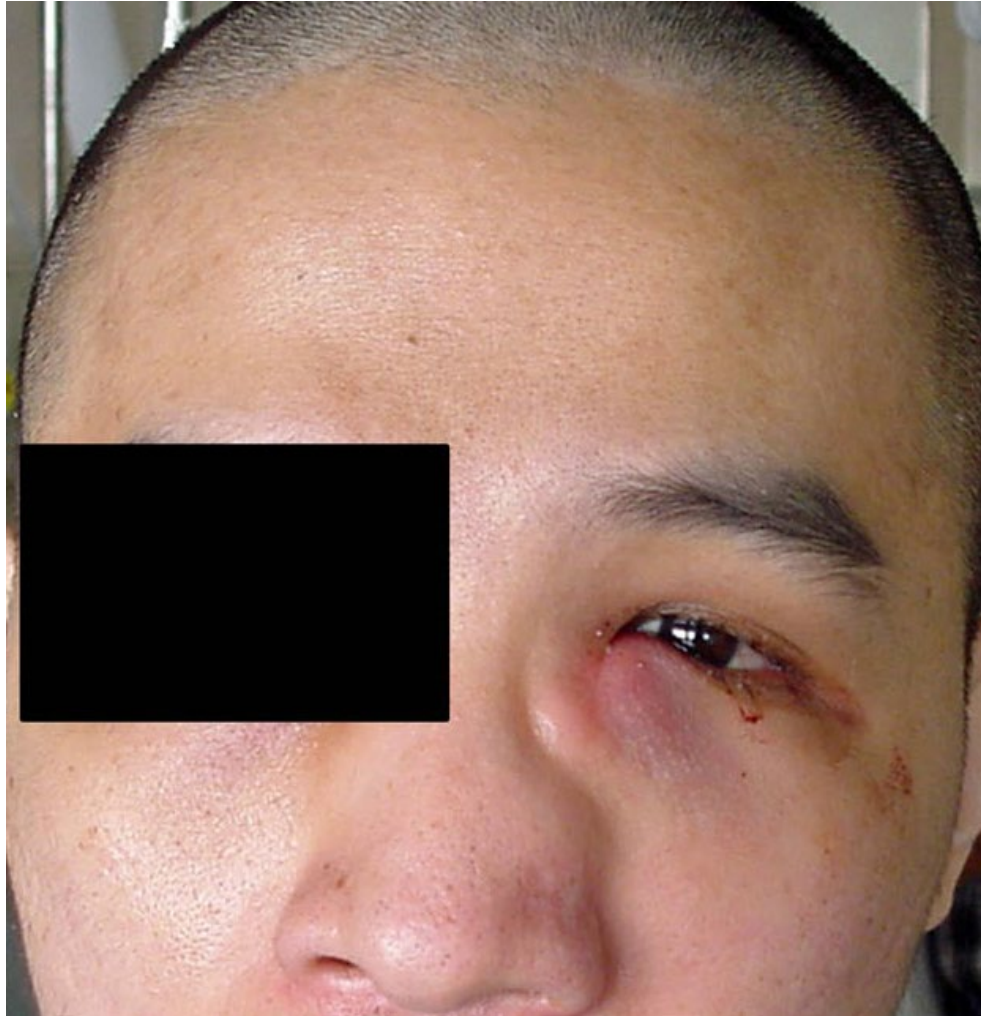
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- CLL/SLL is the most prevalent lymphoid malignancy in North America and Europe and is less common among people of African or Asian origin
- Median age 72 years
- M:F = 2:1
- CLL, including the SLL variant, is usually diagnosed on evaluation of an incidental finding of asymptomatic
- Leukocytosis/lymphocytosis or lymphadenopathy/splenomegaly.
- Only ~20% of patients have symptomatic disease at diagnosis.
- CLL can present with symptomatic anemia, bleeding due to thrombocytopenia, symptomatic adenopathy or splenomegaly (abdominal distention or early satiety), or constitutional symptoms.
- Binet staging / Rai Staging

Extranodal NK/T cell lymphoma



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NK/T-cell lymphomas



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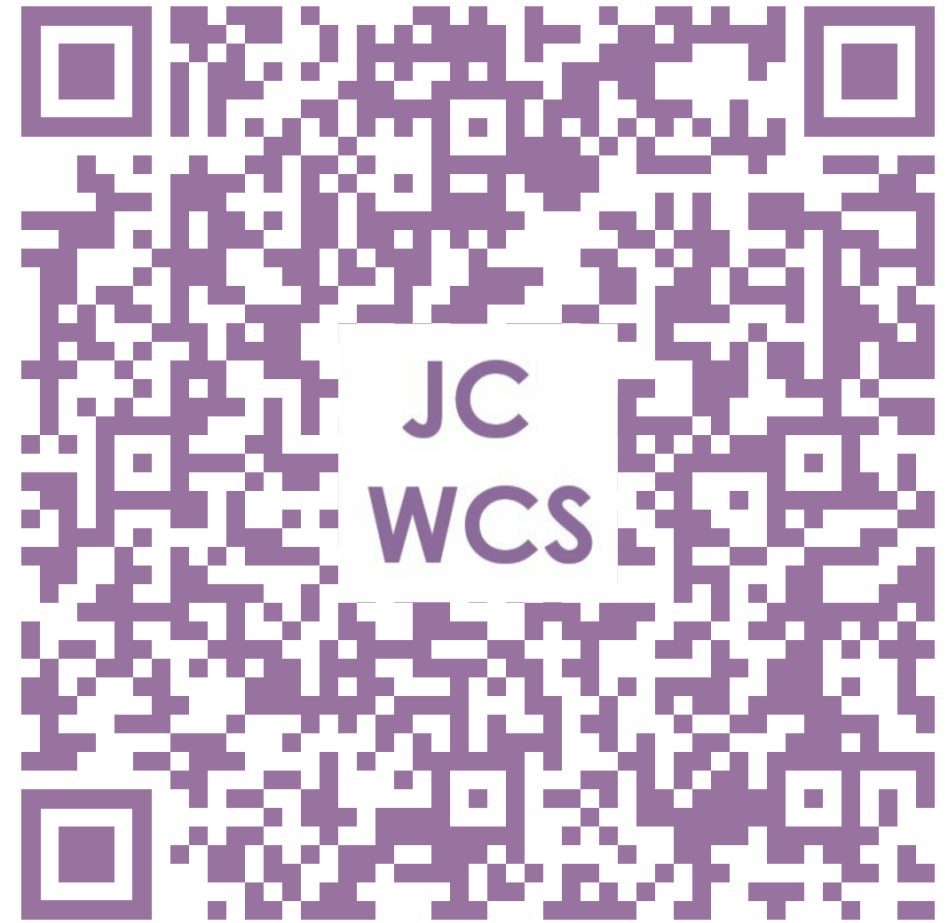
- Geographical predilection for Asian and South American populations and rare in other countries
- Commonly a locally invasive disease destroying nasal and midline facial structures
- Other non-nasal sites that may be involved include skin, GI tract, CNS and testes
- Primary nodal presentation is rare
- Aggressive NK-cell leukaemia/lymphoma is a rare presentation with fever, rash, hepatosplenomegaly, pancytopenia and hemophagocytosis
- An EBV-driven lymphoma
- Treatment involves multi-agent chemotherapy comprising steroids, ifosfamide, cisplatin, L-asparaginase, etoposide

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!



Look for today's teacher
and the others you may have missed.
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